

OPNsense Firewall Rule Configuration and Verification Walkthrough

Firewall Rule Configuration, Testing and Logging Process

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Introduction

This report focuses on the firewall configuration, testing, and verification phase of the OPNsense lab. It includes firewall rule creation, alias configuration, connectivity testing, rule priority verification, and firewall log analysis.

1. Renaming the LAN Interface to Homelab

On the Interfaces: [LAN] page, the interface Description field is changed from the default label to "Homelab" while the underlying device em1 and the existing static IPv4 configuration at 192.168.230.2 remain unchanged. Enable Interface stays checked, and Block private networks and Block bogon networks are left unchecked so that lab traffic between the Kali and Windows clients is not restricted at the interface level.

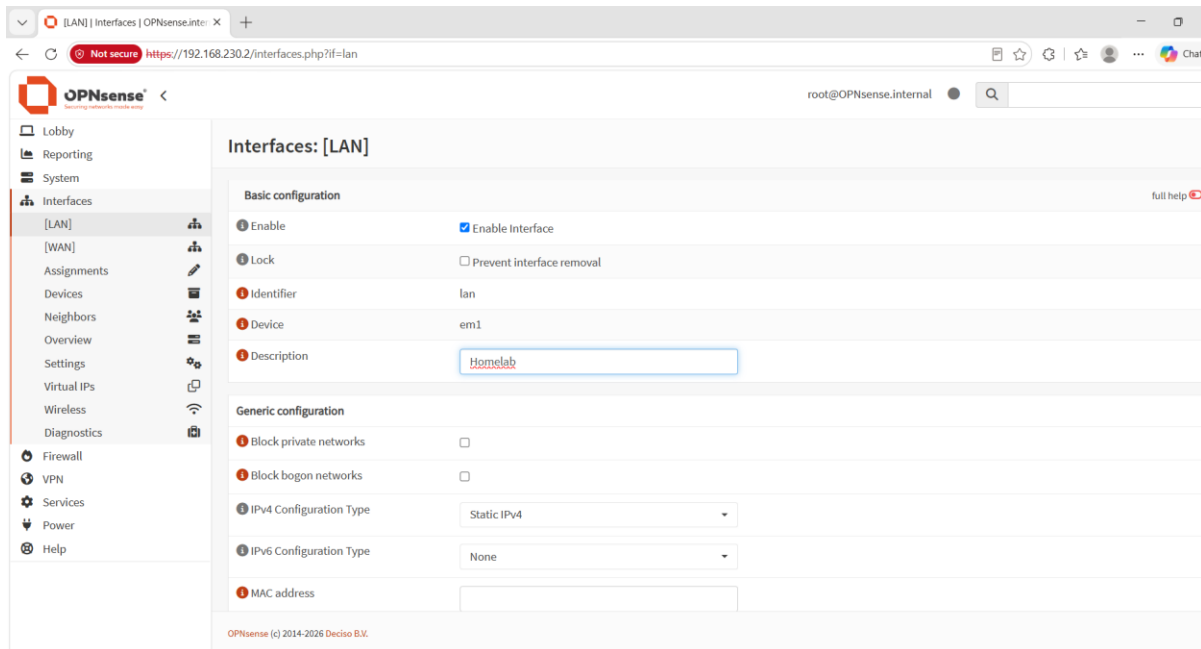


Figure 1: Renaming the LAN Interface to Homelab

2. Ping Connectivity Test from Kali Linux

From the Kali Linux client, ping 192.168.230.2 confirms reachability of the OPNsense LAN gateway, with two replies at 2.76 ms and 5.02 ms round-trip time and 0% packet loss. A second test, ping 8.8.8.8, confirms outbound internet connectivity through the firewall, with a reply received at 239 ms.

```
(kali㉿kali)-[~]
$ ping 192.168.230.2
PING 192.168.230.2 (192.168.230.2) 56(84) bytes of data.
64 bytes from 192.168.230.2: icmp_seq=1 ttl=64 time=5.02 ms
64 bytes from 192.168.230.2: icmp_seq=2 ttl=64 time=2.76 ms
^C
— 192.168.230.2 ping statistics —
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 2.761/3.891/5.022/1.130 ms

(kali㉿kali)-[~]
$ ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=116 time=239 ms
^C
— 8.8.8.8 ping statistics —
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 239.778/239.778/239.778/0.000 ms
```

Figure 2: Kali Linux Ping Connectivity Test

3. Ping Connectivity Test from Windows

The same connectivity checks are repeated from a Windows client using the command prompt. Ping 192.168.230.2 returns two replies with 0% loss and an average round-trip time of 5 ms, and ping 8.8.8.8 also returns two replies with 0% loss, confirming that both the Kali and Windows clients on the Homelab network can reach the gateway and the internet before any firewall rules are added.

```
>ping 192.168.230.2

Pinging 192.168.230.2 with 32 bytes of data:
Reply from 192.168.230.2: bytes=32 time=7ms TTL=64
Reply from 192.168.230.2: bytes=32 time=3ms TTL=64

Ping statistics for 192.168.230.2:
    Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 7ms, Average = 5ms
Control-C
^C
C:\Users\>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=60ms TTL=117
Reply from 8.8.8.8: bytes=32 time=119ms TTL=117

Ping statistics for 8.8.8.8:
    Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 60ms, Maximum = 119ms, Average = 89ms
Control-C
^C
```

Figure 3: Windows Ping Connectivity Test

4. Firewall Rules Before Adding the Block Rule

The Firewall: Rules: Homelab page currently lists only the automatically generated default rules: "Default allow LAN to any rule" for IPv4 and "Default allow LAN IPv6 to any rule" for IPv6, both matching Homelab net as the source and any destination. The Add button in the rule list toolbar is used to begin creating a new custom rule.

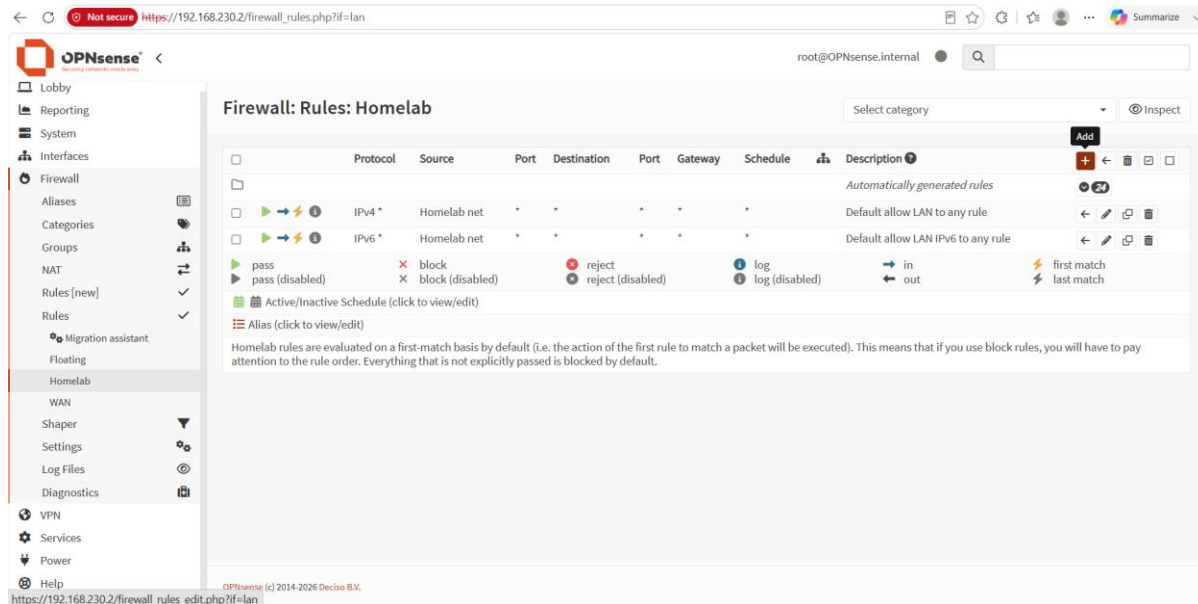


Figure 4: Firewall Rules Before Adding the Block Rule

5. Adding the ICMP Block Rule (Before Apply)

A new rule named "Block icmp to google dns" is added with TCP/IP version IPv4, protocol ICMP, source Homelab net and destination 8.8.8.8. Once saved, the rule list displays a blue notice stating that the firewall rule configuration has been changed and must be applied, along with an Apply changes button.

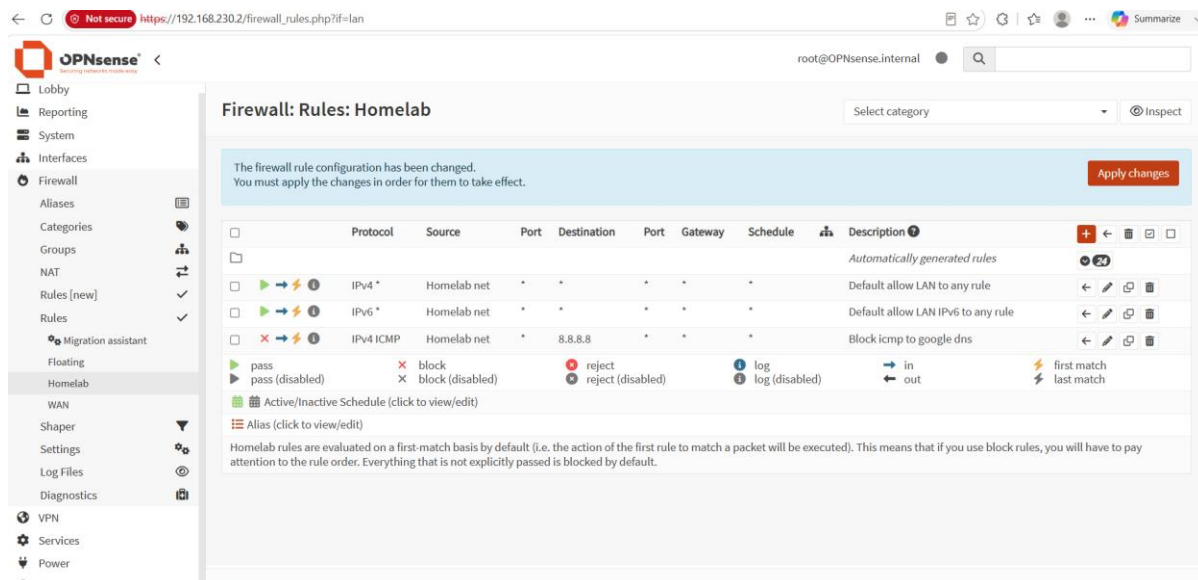


Figure 5: ICMP Block Rule Added Before Apply

6. ICMP Block Rule Applied Successfully

Clicking Apply changes produces the confirmation message "The changes have been applied successfully." The rule order now places "Block icmp to google dns" (IPv4 ICMP, Homelab net to 8.8.8.8) above the two default allow rules, which matters because Homelab rules are evaluated on a first-match basis.

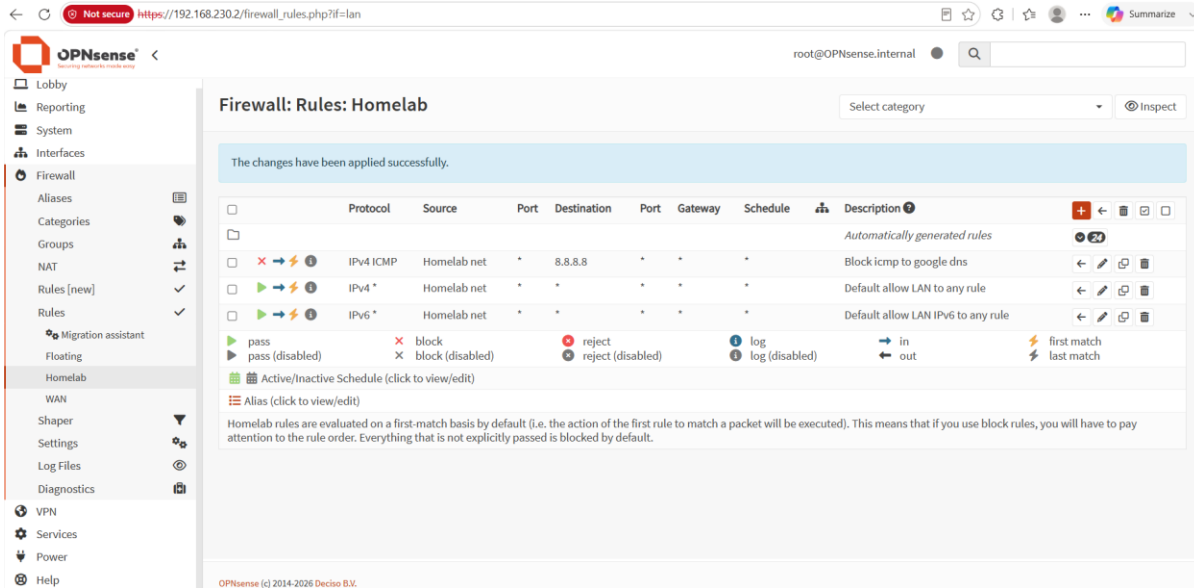


Figure 6: ICMP Block Rule Applied Successfully

7. Firewall Aliases Overview

The Firewall: Aliases page lists the aliases available on the system, including the External (advanced) aliases bogons, bogonsv6, sshlockout and virusprot, alongside the Internal (automatic) aliases __lan_network, __lo0_network and __wan_network. This page is used as the starting point for creating a custom alias to group the Google DNS addresses together.

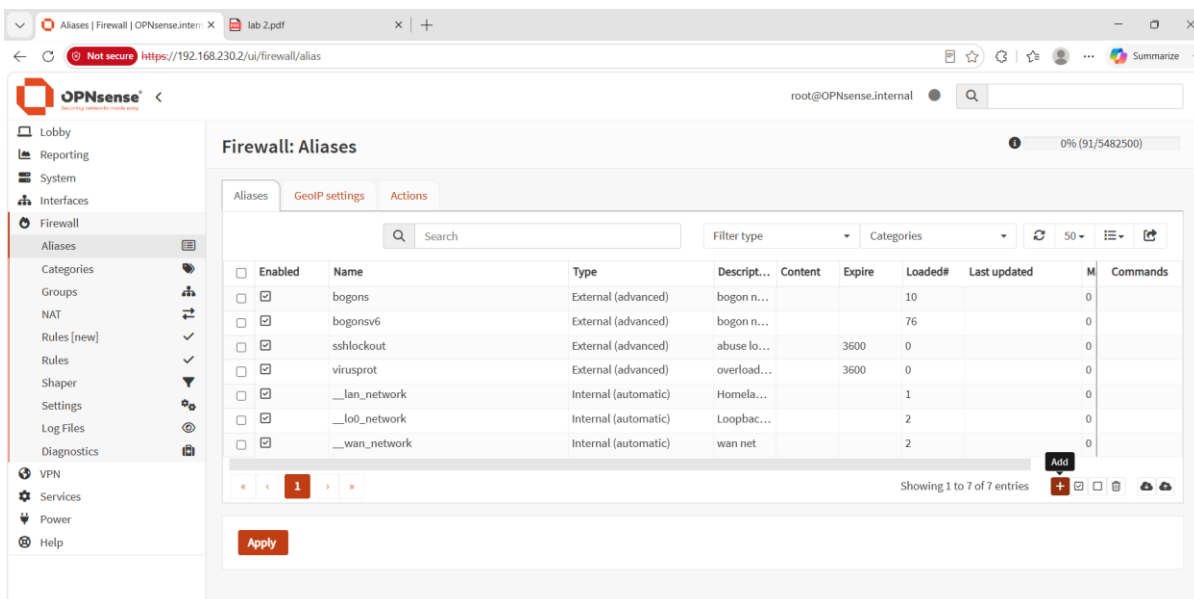


Figure 7: Firewall Aliases Overview

8. Configuring the ICMP Block Rule Using the GoogleDNS Alias

The existing block rule is edited so that the single literal IP address is replaced with an alias-based destination. The Protocol is kept as ICMP with ICMP type Any, the Source remains Homelab network, and the Destination is changed to the GoogleDNS alias, which groups the 8.8.8.8 and 8.8.4.4 addresses under one reusable object.

The screenshot shows the OPNsense Firewall Rule configuration interface. The left sidebar contains navigation links: system, interfaces, firewall, aliases, categories, groups, NAT, Rules [new], Rules, Migration assistant, Floating, Homelab, WAN, Shaper, Settings, Log Files, Diagnostics, VPN, services, power, and help. The main configuration area is for the 'Homelab' rule. The 'TCP/IP Version' is set to 'IPv4'. The 'Protocol' is 'ICMP' and the 'ICMP type' is 'Any'. The 'Source / Invert' section has 'Use this option to invert the sense of the match.' unchecked, and the 'Source' is 'Homelab network'. The 'Destination / Invert' section also has 'Use this option to invert the sense of the match.' unchecked, and the 'Destination' is 'GoogleDNS'. The 'Destination port range' section has 'from:' and 'to:' both set to 'any'. The 'Log' checkbox is unchecked, and the 'Category' field is empty. The footer shows 'OPNsense (c) 2014-2026 Deciso B.V.'.

Figure 8: Configuring the ICMP Block Rule Using the GoogleDNS Alias

9. ICMP Block Rule Added Using the GoogleDNS Alias

After saving, the Homelab rule list shows "Block icmp to google dns" with its Destination column now displaying the linked GoogleDNS alias instead of a plain IP address. The rule still sits above the two default allow rules, so ICMP traffic from the Homelab network toward any address contained in the alias continues to be blocked first.

The screenshot shows the 'Firewall: Rules: Homelab' page. The table lists the following rules:

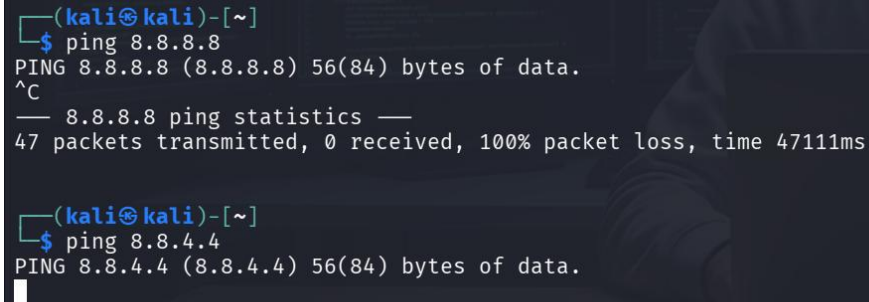
	Protocol	Source	Port	Destination	Port	Gateway	Schedule	Description
Automatically generated rules								
☐	IPv4 ICMP	Homelab net	*	GoogleDNS	*	*	*	Block icmp to google dns
☐	IPv4 *	Homelab net	*	*	*	*	*	Default allow LAN to any rule
☐	IPv6 *	Homelab net	*	*	*	*	*	Default allow LAN IPv6 to any rule

Below the table, there are sections for 'pass' and 'block' rules, 'Active/Inactive Schedule', and an 'Alias' section. A note at the bottom states: 'Homelab rules are evaluated on a first-match basis by default (i.e. the action of the first rule to match a packet will be executed). This means that if you use block rules, you will have to pay attention to the rule order. Everything that is not explicitly passed is blocked by default.'

Figure 9: ICMP Block Rule Added Using the GoogleDNS Alias

10. Testing the ICMP Block Rule Against the GoogleDNS Alias

Pinging 8.8.8.8 from Kali results in 47 packets transmitted and 0 received, a 100% packet loss, confirming the address is blocked. A follow-up ping to 8.8.4.4 shows the same lack of replies, verifying that both addresses grouped inside the GoogleDNS alias are blocked by the single ICMP rule.



```
(kali㉿kali)-[~]  
$ ping 8.8.8.8  
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.  
^C  
— 8.8.8.8 ping statistics —  
47 packets transmitted, 0 received, 100% packet loss, time 47111ms  
  
(kali㉿kali)-[~]  
$ ping 8.8.4.4  
PING 8.8.4.4 (8.8.4.4) 56(84) bytes of data.  
■
```

Figure 10: Testing the ICMP Block Rule Against Both GoogleDNS Alias Addresses

11. Creating an Allow-All Firewall Rule

A new rule is created with Action set to Pass, Interface Homelab, Direction in, TCP/IP version IPv4 and Protocol any, so it is not limited to ICMP traffic. The Source is set to Homelab network, the Destination is left as any, and the Description field is set to "Allow all" to identify the rule's purpose in the rule list.

Firewall: Rules: Homelab

Edit Firewall rule

1 Action	Pass
1 Disabled	☐ Disable this rule
1 Quick	☒ Apply the action immediately on match.
1 Interface	Homelab
1 Direction	in
1 TCP/IP Version	IPv4
1 Protocol	any
1 Source / Invert	☐ Use this option to invert the sense of the match.
1 Source	Homelab network
	Advanced
1 Destination / Invert	☐ Use this option to invert the sense of the match.
1 Destination	any
1 Destination port range	from: any
1 Log	☐ Log packets that are handled by this rule
1 Category	
1 Description	Allow all
1 No XMLRPC Sync	☐

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Figure 11: Creating an Allow-All Firewall Rule

12. Ping Restored After Placing the Allow-All Rule Above the Block Rule

With the new Allow all rule positioned above the ICMP block rule in the Homelab rule list, pinging 8.8.8.8 and 8.8.4.4 from Kali both succeed with 0% packet loss. This demonstrates that because Homelab rules are processed on a first-match basis, a passing rule placed above a blocking rule takes effect first and overrides the block for matching traffic.

```
(kali㉿kali)-[~]
$ ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=116 time=127 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=116 time=122 ms
^C
— 8.8.8.8 ping statistics —
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 121.534/124.367/127.200/2.833 ms

(kali㉿kali)-[~]
$ ping 8.8.4.4
PING 8.8.4.4 (8.8.4.4) 56(84) bytes of data.
64 bytes from 8.8.4.4: icmp_seq=1 ttl=116 time=105 ms
64 bytes from 8.8.4.4: icmp_seq=2 ttl=116 time=103 ms
^C
— 8.8.4.4 ping statistics —
2 packets transmitted, 2 received, 0% packet loss, time 1001ms
rtt min/avg/max/mdev = 103.499/104.415/105.331/0.916 ms
```

Figure 12: Ping Restored After Placing the Allow-All Rule Above the Block Rule

13. Restoring the Block Rule to the Top of the Rule Set

The rule order is adjusted again so that "Block icmp to google dns", with its destination set to the GoogleDNS alias, is moved back above the Allow all rule, followed by the two default allow rules. This restores the ICMP block as the first matching rule for traffic destined to the addresses contained in the GoogleDNS alias.

Firewall: Rules: Homelab

The changes have been applied successfully.

	Protocol	Source	Port	Destination	Port	Gateway	Schedule	Description
Automatically generated rules								
☐	IPv4 ICMP	Homelab net	*	GoogleDNS	*	*	*	Block icmp to google dns
☐	IPv4 *	Homelab net	*	*	*	*	*	Allow all
☐	IPv4 *	Homelab net	*	*	*	*	*	Default allow LAN to any rule
☐	IPv6 *	Homelab net	*	*	*	*	*	Default allow LAN IPv6 to any rule

pass (disabled) block (disabled) reject (disabled) log (disabled) in out first match last match

Active/Inactive Schedule (click to view/edit)

Alias (click to view/edit)

Homelab rules are evaluated on a first-match basis by default (i.e. the action of the first rule to match a packet will be executed). This means that if you use block rules, you will have to pay attention to the rule order. Everything that is not explicitly passed is blocked by default.

Figure 13: Restoring the Block Rule to the Top of the Rule Set

14. Disabling the GoogleDNS Block Rule

To confirm that the block rule itself is responsible for the loss of connectivity, the "Block icmp to google dns" rule is disabled while the Allow all rule and the default allow rules remain active. With the block rule disabled, only the passing rules are left to evaluate matching Homelab traffic.

☐		Protocol	Source	Port	Destination	Port	Gateway	Schedule
📁								
☐		IPv4 ICMP	Homelab net	*	GoogleDNS	*	*	*
☐		IPv4 *	Homelab net	*	*	*	*	*
☐		IPv4 *	Homelab net	*	*	*	*	*
☐		IPv6 *	Homelab net	*	*	*	*	*
	pass		block		reject		log	
	pass (disabled)		block (disabled)		reject (disabled)		log (disabled)	
	Active/Inactive Schedule (click to view/edit)							
	Alias (click to view/edit)							
Homelab rules are evaluated on a first-match basis by default (i.e. the action of the first rule to match a packet will be executed according to the rule order. Everything that is not explicitly passed is blocked by default.								

Figure 14: Disabling the GoogleDNS Block Rule

15. Ping Successful After Disabling the Block Rule

With the ICMP block rule disabled, pinging 8.8.8.8 returns two replies with 0% packet loss and round-trip times around 106-112 ms, and pinging 8.8.4.4 also returns two replies with 0% packet loss and round-trip times around 51-63 ms. Both results confirm that disabling the block rule allows the Allow all rule to permit ICMP traffic to both Google DNS addresses.

```
(kali㉿kali)-[~]
$ ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=116 time=106 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=116 time=112 ms
^C
— 8.8.8.8 ping statistics —
2 packets transmitted, 2 received, 0% packet loss, time 1005ms
rtt min/avg/max/mdev = 106.415/108.990/111.565/2.575 ms

(kali㉿kali)-[~]
$ ping 8.8.4.4
PING 8.8.4.4 (8.8.4.4) 56(84) bytes of data.
64 bytes from 8.8.4.4: icmp_seq=1 ttl=116 time=51.8 ms
64 bytes from 8.8.4.4: icmp_seq=2 ttl=116 time=63.3 ms
^C
— 8.8.4.4 ping statistics —
2 packets transmitted, 2 received, 0% packet loss, time 1003ms
rtt min/avg/max/mdev = 51.831/57.547/63.263/5.716 ms
```

Figure 15: Ping Successful After Disabling the Block Rule

16. Ping Blocked Again After Re-Enabling the Block Rule

After the "Block icmp to google dns" rule is re-enabled and applied, the same ping tests are repeated. Ping 8.8.8.8 now shows 3 packets transmitted and 0 received with 100% packet loss, and ping 8.8.4.4 shows the same result, confirming that the ICMP block rule is once again the first rule to match and drop traffic to the GoogleDNS alias addresses.

```
rtt min/avg/max/mdev = 51.831/57.547/63.263/5.716 ms

(kali㉿kali)-[~]
$ 
(kali㉿kali)-[~]
$ ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
^C
— 8.8.8.8 ping statistics —
3 packets transmitted, 0 received, 100% packet loss, time 2056ms

(kali㉿kali)-[~]
$ ping 8.8.4.4
PING 8.8.4.4 (8.8.4.4) 56(84) bytes of data.
^C
— 8.8.4.4 ping statistics —
3 packets transmitted, 0 received, 100% packet loss, time 2030ms

(kali㉿kali)-[~]
```

Figure 16: Ping Blocked Again After Re-Enabling the Block Rule

17. Enabling Logging on the ICMP Block Rule

The "Block icmp to google dns" rule is edited once more, this time to check the Log packets that are handled by this rule option while keeping the Destination set to the GoogleDNS alias. Enabling logging on this rule allows every ICMP packet it blocks to be recorded in the firewall log for later review.

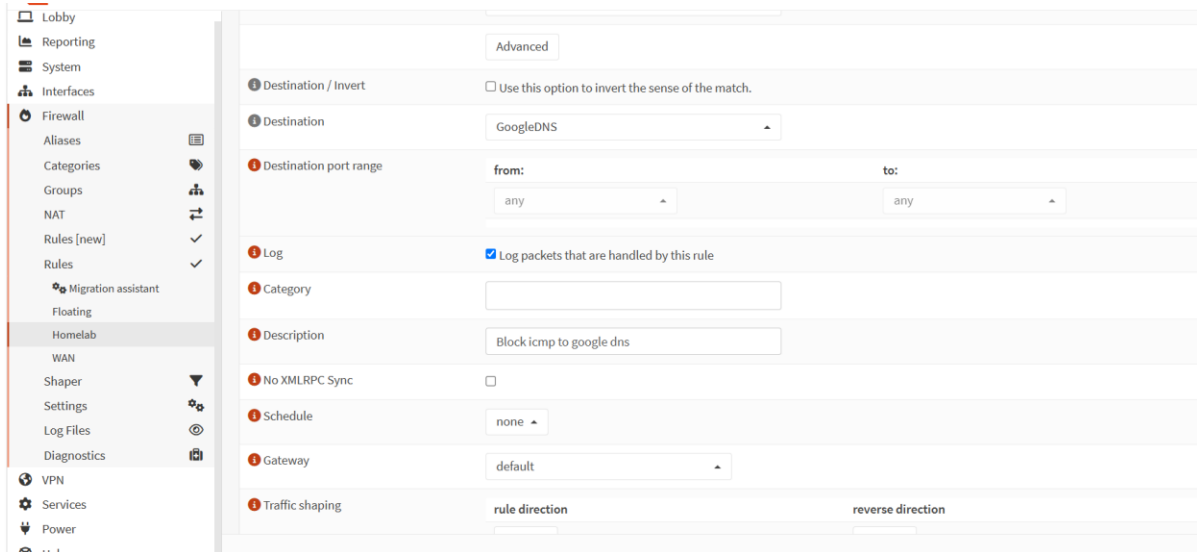


Figure 17: Enabling Logging on the ICMP Block Rule

18. Generating Ping Traffic for Log Verification

With logging enabled on the block rule, ping 8.8.8.8 is run again from the Kali client to generate a stream of blocked ICMP packets. This traffic is used in the next step to confirm that the logging option correctly records the blocked packets in the OPNsense firewall log.

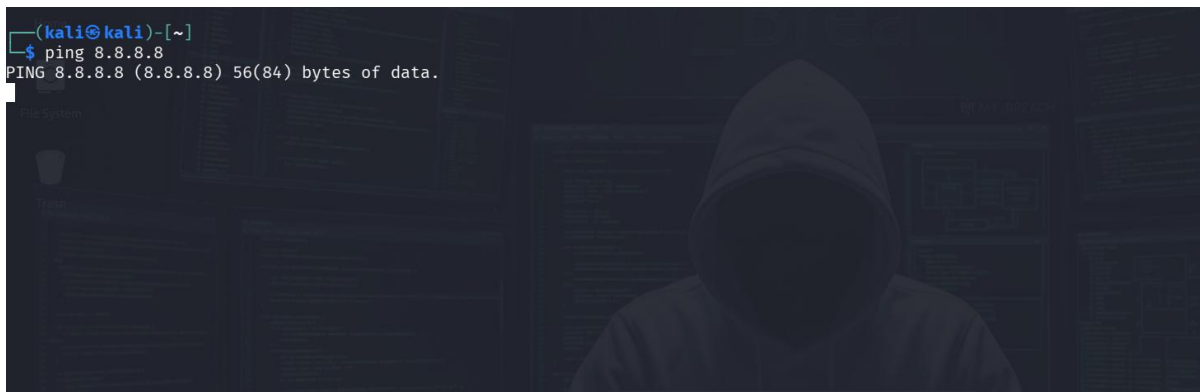


Figure 18: Generating Ping Traffic for Log Verification

19. Firewall Live Log Showing Blocked ICMP Packets

The Firewall: Log Files: Live View page displays the resulting log entries, each showing Interface Homelab, direction In, protocol ICMP, source 192.168.230.99 and destination 8.8.8.8, with the action column marked block and the label "Block icmp to google dns". The repeated, timestamped entries confirm that the ICMP block rule with logging enabled is actively capturing every blocked packet sent toward the GoogleDNS alias.

The screenshot shows the OPNsense Firewall: Log Files: Live View page. The left sidebar contains navigation links: Lobby, Reporting, System, Interfaces, Firewall (selected), Aliases, Categories, Groups, NAT, Rules [new], Rules, Shaper, Settings, Log Files, General, Live View (selected), Overview, Plain View, Diagnostics, VPN, Services, Power, and Help. The main content area displays a table of log entries. The table has columns: Interface, Time, Protocol, Source, Destination, Action, and Label. The entries show multiple blocked ICMP packets from 192.168.230.99 to 8.8.8.8. The table is filtered by 'action' containing 'block'. The 'Active filters' section shows 'action' and 'contains' with a search box. The 'Options' section includes 'Auto-refresh' (checked), 'Lookup hostnames' (unchecked), and 'Select any of given criteria (or)' (unchecked). The 'Table size' is set to 25 and the 'History size' is set to 10000. The footer indicates 'OPNsense (c) 2014-2026 Deciso B.V.'.

Interface	Time	Protocol	Source	Destination	Action	Label
Homelab	In 2026-07-02T00:51:29	ICMP	192.168.230.99	8.8.8.8	block	Block icmp to google dns
Homelab	In 2026-07-02T00:51:28	ICMP	192.168.230.99	8.8.8.8	block	Block icmp to google dns
Homelab	In 2026-07-02T00:51:27	ICMP	192.168.230.99	8.8.8.8	block	Block icmp to google dns
Homelab	In 2026-07-02T00:51:26	ICMP	192.168.230.99	8.8.8.8	block	Block icmp to google dns
Homelab	In 2026-07-02T00:51:25	ICMP	192.168.230.99	8.8.8.8	block	Block icmp to google dns
Homelab	In 2026-07-02T00:51:24	ICMP	192.168.230.99	8.8.8.8	block	Block icmp to google dns
Homelab	In 2026-07-02T00:51:23	ICMP	192.168.230.99	8.8.8.8	block	Block icmp to google dns
Homelab	In 2026-07-02T00:51:22	ICMP	192.168.230.99	8.8.8.8	block	Block icmp to google dns
Homelab	In 2026-07-02T00:51:21	ICMP	192.168.230.99	8.8.8.8	block	Block icmp to google dns
Homelab	In 2026-07-02T00:51:20	ICMP	192.168.230.99	8.8.8.8	block	Block icmp to google dns

Figure 19: Firewall Live Log Showing Blocked ICMP Packets

Conclusion

This report documented the firewall rule configuration, testing, and verification phase of the OPNsense homelab. After renaming the LAN interface to Homelab and confirming baseline connectivity from both Kali Linux and Windows clients, a custom ICMP rule was created to block traffic to Google's public DNS servers, first using a single literal IP address and then using a reusable GoogleDNS alias grouping 8.8.8.8 and 8.8.4.4.

Testing with an Allow all rule, and by enabling and disabling the block rule, demonstrated how OPNsense evaluates Homelab rules on a first-match basis, so rule order directly determines whether traffic is passed or blocked. Finally, enabling the Log packets option on the ICMP block rule and reviewing the Firewall: Log Files: Live View page confirmed that every blocked packet toward the GoogleDNS alias was correctly captured with its interface, protocol, source, destination and action, verifying that the firewall rules, aliases and logging were all configured and functioning as intended.